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WHEN SRINATH RAVICHANDRAN, 36, and Moin S.P.M., 31, aren't hitting the ball across the boundary in a game of backyard cricket, they are working on sending their Agnibaan series of low-cost rockets into space. Ravichandran, who has a master's degree in aerospace engineering from the University of Illinois, Urbana-Champaign, first thought of the idea while networking with people from the commercial space domain in Los Angeles who had designed satellites but didn't know how to put them into the earth's orbit. "When I naively asked around as to why that was a problem, I was told that it was due to either the absence of the right kind of rockets or partners," he tells *Business Today*. Finally, Ravichandran, who had previously worked as an electrical engineer and a finance person, and was tracking the development of Elon Musk's SpaceX programme, quit his job and returned to India. Here, he co-founded Agnikul Cosmos with Moin—they had bonded over friendly cricket matches at the same ground in Chennai—in 2017 to be a part of what some commentators have termed the new space rage.

The duo is part of an ever-increasing tribe of new-age entrepreneurs who have consciously chosen to foray into the commercial space sector, a territory where not many in the country had dared to venture earlier. The

final frontier is the new rage among several start-ups that are working on everything and anything to do with space, that is, from building launch systems and satellites to working on other related applications.

The commercial space industry holds the promise of emerging as the new sunrise sector, with the central government projecting it as contributing 1 per cent to the country's \$5-trillion economy target by 2024. Already, with 368 private firms, India's space economy is the fifth-largest in the world after the US, the UK, Canada and Germany, says a report by SpaceTech Analytics. More than 75 start-ups are registered under the space technology category on the Startup India portal, according to government data. Moreover, several more start-ups registered under other categories on the same portal are also involved in the space domain.

Digantara Research & Technologies, founded by Anirudh N. Sharma, Rahul Rawat and Tanveer Ahmed, is looking at launching a constellation of satellites equipped with state-of-the-art payloads to scan and map the low-Earth environment. They also have plans to build the world's largest catalogue of man-made objects in space, including those as small as 1 cm, for better operational and situational awareness in space. Sharma, 23, and Rawat, 22, became entrepreneurs in 2018 after they successfully developed a satellite structure for a foreign space agency and were paid about \$1,000 for it. They were studying computer science engineering at Lovely Professional University (LPU), Phagwara, at the time. As active members of the university's student satellite programme team, they had already contributed to building a nano satellite with the Indian Space Research Organisation (ISRO) and participated in a couple of international conferences.

Sharma used to write cold emails to space agencies around the world seeking assistance. And that is how he got connected with the agency in question, which started working with them at the university level. "It was a great learning curve for us from 2018 to 2020, where we learnt a lot about business and fundraising. When you are building a company that intends to solve the problem of space junk, who do you think will agree to invest money in something like that? It's a difficult proposition that we were offering," says Sharma with a smile. "We withdrew the money meant to pay our college fee to fund our first contract as component manufacturers," adds Rawat, laughing.

\$7

BILLION

**INDIA'S SHARE
OF THE GLOBAL
COMMERCIAL
SPACE SECTOR**

\$360

BILLION

**SIZE OF THE
COMMERCIAL SPACE
SECTOR GLOBALLY**